Project C: Portfolio and Risk Management

1. Performance modeling: You are a fund-of-funds manager evaluating a potential investment in a hedge fund. It is the end of 2006, and you have in-depth access to the fund's daily position data for the period 2004-2006. Your goal is to characterize the past return profile in order to estimate its future success. Use the data for the equity long/short strategy called "Contra01," which is available in the OLAP database **StatArb03,** using the data cube **Positions**. For performance results, use the weighted returns net of transaction costs, given by the fields "Weighted Return Tc." Ignore hedge fund management fees.

- **a.** Give the annualized return, volatility, and Sharpe ratio for the strategy.
- **b.** Use ordinary least-squares linear regression to report on the CAPM alpha, beta, and *R*-squared for the strategy, including standard errors of your estimates and *t*-statistics. Is the alpha statistically different from a CAPM prediction? How would you characterize the market-neutrality of the fund in light of its beta?
- **c.** Extend this regression model to include Fama-French factors, and repeat the analysis in the previous question. (Data is available in crispy09.dbo.famafrench.)
- d. Plot the strategy one-day returns in ranked order, from lowest to highest. What fraction of days were winners, and what fraction were losers? What was the median return of winners, and what was the median return of the losers?
- **e.** Based on your analysis in 2006, how do you expect the fund to perform for the three-year period ahead? Do you have any other perspectives on how to predict the future, out-of-sample, investment performance?

2. Performance attribution: Analyze the performance of the contrarian equity long/short strategy "Contra01" over the period 2007-2009. Use the data in the OLAP database **StatArb03,** using the cube **Positions**. For performance results, use the weighted returns net of transaction costs, given by the fields "Weighted Return Tc"

- a. Give the annualized return, volatility, and Sharpe ratio for the strategy overall. Do the same for the long and short halves (or sub-portfolios) of the strategy. (That is, evaluate the same performance analytics just using the long positions, and then just the short positions.)
- **b.** Use linear regression to report on the CAPM alpha, beta, and *R*-squared for the strategy, including standard errors of your estimates and *t*-statistics. Is the alpha statistically different from a CAPM prediction? How would you characterize the market-neutrality of the fund in light of its beta?
- **c.** Best winner and worst loser: What was the best and the worst single-day performance of an individual position? That is, for which stock held in the portfolio, and on which day, was its contribution to the portfolio return the highest or the lowest? What was each stock's simple return, and what was its portfolio weight?
- d. Distribution of winners and losers: Plot the strategy one-day returns in order, from lowest to highest. What fraction of days were winners and what fraction were losers? What was the median return of winners and what was the median return of the losers?

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e. How do all these results compare with your predictions from the period 2004-2006 from Question #1 above? Did your Q1 predictions above in fact predict the subsequent, out-of-sample, performance? Do you believe the portfolio manager's style remained constant over time, or did it change? Why or why not?
f. Which SIC sector contributed the largest total return, in absolute value, for calendar year 2009? What was the return, and what fraction of strategy total return did it contribute? (A plot or table of total weighted return, aggregated by sector and filtered on 2009 tradedates would be helpful.)
g. Same as previous question using GICS sectors.

3. Risk measurement: Analyze the exposures of the contrarian equity long/short strategy "Contra01" using the fund data for 2007-2009 in the OLAP database **StatArb03** using the cube **Positions**. This strategy focused on security selection and mean-reversion dynamics at the individual stock level. While the overall longs and shorts were constrained to be balanced, the individual sectors and other factors were not. Use the data on the history of portfolio exposures, measured primarily by aggregating security weights, to answer the following.

a. For each GICS sector, give its highest and lowest net exposure over the period 2007-2009. Which sectors stayed within +/- 10% of net portfolio weight throughout?
b. On 9/12/2008, which GICS sector was the most unbalanced? Give the total portfolio weight long, total weight short, and net portfolio weight for that sector. What was the day's portfolio return?
c. On 9/15/2008, which stock contributed the largest absolute value of return to the portfolio? What was the impact? Would you consider it an outlier? Why or why not?
d. Suppose a sub-strategy were defined by restricting the "Contra01" portfolio weights and returns to the GICS Financials sector only. Is this sub-strategy "market-neutral" on its own? How do the sub-strategy exposures evolve over time in the absence of constraints? Plot and the net long/short exposure as well as the gross long vs. gross short exposures.

4. Correlation dynamics: Analyze the behavior of the *average correlation* among stocks in the Dow Jones Industrial Average using the data in **crispy09.** (You may find it helpful to use the tables and views **indexmember, fact_equity, view_fact_index**, **view_fact_indexmember,** and to consider how the views are constructed.)

Create two plots: one for the period 2008-2009 and the second using the period 2019-2020. On each graph, plot lines for the following 4 daily time series. For each analytic that requires and observation window, use a trailing 3-month (63-day) period.

- Average ("implied") correlation, as discussed in class,

$$\overline{\rho} \equiv \frac{\sigma_p^2 - \sum_{i=1}^{N}(w_i\sigma_i)^2}{2\sum_{i<j}(w_i\sigma_i)(w_j\sigma_j)}$$

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- Index realized volatility
- Index "volatility$_0$": σ_0 is the volatility the index would have if all member correlations were zero (i.e., square root of σ_0^2, the index "variance$_0$" discussed in class)
- Index "volatility$_1$": similarly, σ_1 is the volatility the index would have if all member correlations were 100%

Answer the following questions:

a. For each period examined, what is the maximum value of the average correlation and when does it occur?
b. What constraints should be satisfied by the four values? Do they hold for the period in question?
c. What could cause the constraints to be violated?

<u>About this project:</u> Each problem in this assignment can be done in multiple ways. It is up to you to select appropriate and efficient tools for each task.

- For example, in Q1, Q2, and Q3, the "Contra01" data can be accessed via the OWC web pages linked on the `hedge` web server, via Excel, or via the Management Studio, while the analytics (vol, Sharpe ratio, etc.) can be done in Excel, in R, Python, Java, C, Matlab, etc. You can have your favorite AI tool write the queries, write the code, write the commentary, or create functions to perform any of these.
- Explain your choices and, as always, document and describe your AI use.
- Benchmark data such as indices, rates, and factor returns can be found, among other places, in the crispy09 tables bench, benchvalue, and famafrench, or you can source them elsewhere.
- For this project only – with the exception of Q2(c) – you may ignore the differences between simple and log returns, assuming (not quite accurately) that they are approximately equal.

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